DualDA-Net: Dual-Head Rectification for Cross-Domain Object Detection of Remote Sensing

Yangguang Zhu, Xian Sun, *Senior Member, IEEE*, Wenhui Diao, *Member, IEEE*, Haoran Wei, *Graduate Student Member, IEEE*, and Kun Fu, *Member, IEEE*

***Abstract*— Object detectors of remote sensing (RS) imagery with deep learning have become increasingly popular and rely heavily on extensive labeled data. The source-only detectors, which are trained on massive labeled data in a source domain, in some cases fail to get satisfactory performance on a target domain due to the domain shift. To alleviate the domain shift, popular approaches consider feature distribution alignment, but the target domain with massive unlabeled data is underutilized. Some methods use the source-only model to generate pseudo-labels for target domain data, but the variation of different RS scenarios produces the domain shift, which is injected directly into the pseudo-labels. Therefore, we propose the dual-head rectification domain adaptation network (DualDA-Net) to alleviate the domain shift and exploit the potential of unlabeled target domain data. DualDA-Net cooperates the coarse-to-fine consistency alignment (CCA) with dual-head co-training (DHCT) to align the distribution and generate pseudo-labels progressively. Specifically, the CCA focuses on source and target domain feature distribution alignment via CCA on multilevel features. Moreover, the DHCT with dual detection heads is deployed in the teacher–student framework, where one of the heads complements the other with high-quality predictions to rectify the pseudo-labels as supervision and alleviate the biased information. Sufficient experiments have been conducted on several domain adaptation settings. The experimental results demonstrate that our DualDA-Net achieves success in the target domain for cross-domain object detection of RS imagery.**

***Index Terms*— Adversarial learning, coarse-to-fine alignment, cross domain, dual-head co-training (DHCT), object detection.**

I. Introduction

OBJECT detection is a fundamental and practical challenge in remote sensing (RS) that involves identifying the location and categorizing objects in images. It has significant applications in building detection, intelligent agriculture, and environmental monitoring [1], [2]. With the advances in RS technology, deep learning algorithms have demonstrated enormous potential in interpreting RS imagery [3], [4], [5], [6], [7], [8], [9]. However, deep learning-based object detection algorithms require extensive labeled data, which can be time-consuming and expensive to obtain. Furthermore, the diverse backgrounds, object scales, and temporal variations in RS scenarios present challenges in data annotation and result in domain shift between different datasets [10]. As a consequence, detectors trained on a source domain dataset with a large amount of labeled data may not perform well on a target domain due to the impact of domain shift. This problem is illustrated in Fig. 1 via 2-D t-stochastic neighbor embedding (t-SNE) [11], where the feature distribution of NWPU VHR-10 [12] and dataset for object detection in aerial images (DOTA) [13] datasets demonstrates the domain shift between different domains. In summary, the problem of domain shift poses a significant challenge to the development of accurate detectors for RS applications.

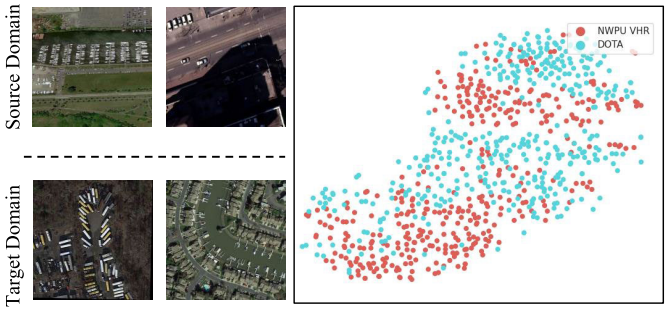


Fig. 1. (Top left and bottom left) Sampled data of the source and target domains, respectively. (Right) Feature visualization via 2-D t-SNE [11] displays the distribution discrepancy between the NWPU VHR-10 [12] and DOTA [13] datasets. Specifically, the features are generated by the model trained on the NWPU VHR-10 [12] dataset.

To improve the performance of deep learning algorithms and mitigate domain shift, unsupervised domain adaptation (UDA) methods [14], [15], [16], [17], [18], [19] have been proposed to align feature distributions across different domains. These methods can be classified as metric-based [14], [15], [20], [21], [22], adversarial-based [16], [23], [24], [25], [26], generative-based [18], [19], [27], and so on. Metric-based algorithms minimize the discrepancy between feature distributions in cross domains using a manually designed

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Yangguang Zhu, Xian Sun, Haoran Wei, and Kun Fu are with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100094, China, also with the Key Laboratory of Network Information System Technology (NIST), Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100190, China, and also with the School of Electronic, Electrical and Communication Engineering, University of Chinese Academy of Sciences, Beijing 100049, China (e-mail: zhuyangguang19@mails.ucas.ac.cn; sunxian@aircas.ac.cn; weihaoran18@mails.ucas.ac.cn; fukun@mail.ie.ac.cn).

Wenhui Diao is with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100094, China, and also with the Key Laboratory of Network Information System Technology (NIST), Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100190, China (e-mail: diaowh@aircas.ac.cn).

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metric loss. With the emergence of the gradient reversal layer (GRL) [28], adversarial-based algorithms use a discriminator to align the source and target domain distributions in an adversarial manner. Generative-based algorithms employ generative adversarial networks (GANs)-based image transfer to generate target-style images from source domain data for model training. These algorithms are currently witnessing increased popularity. Also, some of them [16], [26], [29], [30], [31] are commonly implemented for cross-domain object detection of RS.

Some of the above algorithms [26], [29], [30], [31] achieve excellent performance in cross-domain object detection of RS. Nevertheless, they ignore the potential information in the unlabeled target data, which is in urgent need of being exploited. Recent advances in deep learning have demonstrated the value of unlabeled data, and some algorithms [32], [33], [34], [35], [36] have been developed to use it more effectively. For RS, Zhu et al. [16] have proposed a self-training approach to generate high-quality pseudo-labels for the unlabeled target data, which leverages a model recursively trained with source and target data to generate pseudo-labels. Such a self-training approach can introduce noisy labels biased toward the source domain. The reasons can be outlined as follows. First, the generalization in the target domain is affected by domain shift caused by scenario variations in RS. The model incorporates specific biased information from the source domain. Furthermore, biased information exists within the same categories, such as context differences between instances observed from different perspectives [37], [38]. The model captures features within a conditional probability distribution of instance context, thus implicitly incorporating biased information within the pseudo-labels.

Recently, some approaches [39], [40] have emerged, focusing on alleviating the biased information from the model. For example, Nam et al. [39] propose to simultaneously train two neural networks, one to be biased and the other to be debiased. Such a technique has often been adopted in co-training. Co-training techniques [41], [42], [43], [44] have made progress on unlabeled data, in which they usually concentrate on multinetworks to promote the utilization of unlabeled. Besides, the multiple networks can learn from each other, which can be treated as an essential rectification ability. Inspired by the co-training technique, we adopt a teacher–student framework [34] and propose the dual-head rectification domain adaptation network (DualDA-Net). The DualDA-Net for cross-domain object detection of RS is composed of coarse-to-fine consistency alignment (CCA) and dual-head co-training (DHCT). The existence of domain shift influences the quality of pseudo-labels, thereby damaging the subsequent model training process. For example, the model trained on source domain struggles to incorporate the distribution of instances and scenarios on the limited dataset. However, the model often fails to generalize well in target domain. CCA is deployed in multilevel features extracted by backbone to align the feature distribution. In particular, the CCA focuses on the alignment of spatial semantics in low-level features. Moreover, in the high-level feature extraction stage, CCA strives for the consistency of feature distribution between cross domains

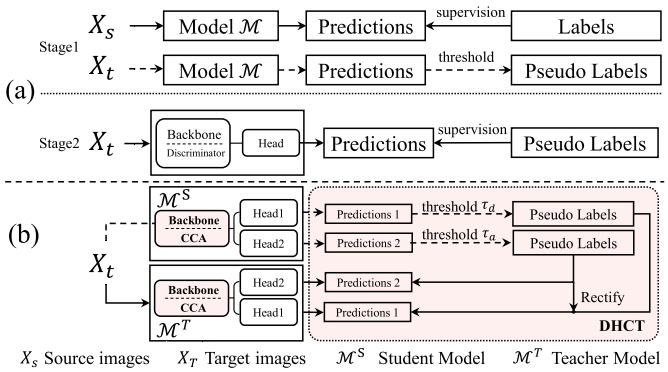


Fig. 2. Comparison with previous methods of domain adaptation in RS. The training process of source images is omitted. (a) Previous methods with self-training. (b) Our proposed Dual-Net utilizes the co-training for cross-domain object detection in RS. Dual heads with different thresholds τ_d and τ_a conduct rectification of pseudo-labels.

by utilizing the dual branches with a gate to align instance information within the channel. Nevertheless, we achieve the co-training strategy with DHCT to utilize the unlabeled target data and reduce the biased information in pseudo-labels. One of the heads is treated as the dominant head, while the other acts as an affiliation of the dominant head. To address any potential low-quality pseudo-labels produced by the dominant head, the affiliated head gradually offers high-quality pseudo-labels while complementing oblivious instances.

The previous methods have developed in the source-only model to directly predict pseudo-labels for the target dataset, as illustrated in Fig. 2. They disregard the impairment caused by biased information in source domain. The refinement of the pseudo-labels is also not considered. To tackle these challenges, we introduce the DualDA-Net method. DualDA-Net employs CCA to extract features and utilizes DHCT with dual heads for pseudo-label rectification. By aligning the distribution of the source and target domains, our DualDA-Net generates more accurate pseudo-labels within the teacher–student framework. Sufficient experiments indicated that our DualDA-Net can significantly boost the efficiency of cross-domain object detection in RS. The following is a summary of our article's significant contributions.

1) A novel DualDA-Net is proposed for UDA object detection in RS. It addresses the biased information in model and domain shift in distribution.
2) The DHCT is designed with dominant and affiliated heads for alleviating the impact of biased information. Besides, to mitigate the effect of domain shift, we innovatively use the CCA for feature alignment on multifeature levels.
3) Extensive experiments have proved the superiority of DualDA-Net on several adaptation settings. Our method achieves excellent results on these UDA object detection in RS.

The subsequent article is structured as follows. In Section II, we discuss some related work. In Section III, we present a comprehensive introduction to our Dual-Net, including its implementation. In Section IV, we delve into our experimental results and conduct a thorough analysis of them. In Section V, we identify and discuss some limitations of our work and

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provide insight into future work. Finally, we summarize our paper and highlight the contributions of our proposed DualDA-Net in Section VI.

II. Related Work

A. Object Detection

Object detection is a critical task that involves locating the instances and categorizing them for the interpretation of RS imagery. Deep learning models have been increasingly utilized for RS object detection, as deep learning techniques have advanced. For object detection, it can be classified into two types based on the type of the predicted box, namely, horizontal object detection [45], [46], [47], [48] and oriented object detection [4], [6], [49], [50], [51]. For horizontal object detection, Fast-region-based convolutional neural networks (RCNN) [45] and Faster-RCNN [46] have emerged and become pioneers in the two-stage horizontal object detection. Subsequently, one-stage methods [52], [53] were proposed to further improve the speed of the model. In addition, some anchor-free methods [48], [54] do not require the presetting of anchors, achieving fast and accurate object localization and classification. Due to the special demands of RS object detection and text detection, oriented object detection [4], [6], [49], [50], [51], [55] has emerged and developed rapidly. Unlike horizontal object detection, oriented object detection mainly captures oriented objects and predicts their minimum bounding rectangles. The existing pipeline for oriented object detection is mainly extended from horizontal object detection. The region of interest (RoI) Transformer [55] is proposed to detect oriented object detection based on the two-stage detector. Deep learning detectors have brought a revolutionary impact on object detection in RS. They enable more accurate and efficient identification of objects in images. However, the effectiveness of these detectors heavily depends on the availability of large labeled datasets. Equally important is the need for similar data distribution between the training and test sets. When there is a domain shift between datasets, achieving satisfactory results from deep learning models becomes difficult.

B. Domain Adaptation

The diversity, variation, and temporal offset of RS scenarios result in inconsistent data distribution between different datasets, which is known as domain shift [10]. The existence of domain shift negatively impacts the performance of cross-domain datasets. To mitigate the effects of domain shift, domain adaptation methods [14], [16], [20], [21], [23], [24], [26], [29] have been developed to improve the generalization of deep learning models. These methods can be categorized into three types: metric-based [14], [15], [20], [21], [22], adversarial-based [16], [23], [24], [25], [26], and generative-based [18], [19], [27]. The most widely used techniques involve aligning the source and target domains before the GRL [28] is established by designing a metric that minimizes the feature embedding distance. For example, Ma et al. [21] have developed a novel domain adaptation classification technique based on the alignment of class centroid and covariance. This method investigates the class centroid and covariance statistics to align the data distribution. Subsequently, the GRL module has been incorporated into the network, and the domain discriminator is designed to align the feature distributions via adversarial loss between the source and target domains. Zhu et al. [23] have proposed a semisupervised center-based discriminative adversarial learning (SCDAL) framework that integrates filtering out easy triplets, a proposed hard triplet loss, and adversarial learning with center loss. In addition, generative-based methods are also popular for domain adaptation, which concentrate on the image-level translation. Based on the CycleGAN model, Soto Vega et al. [27] developed a novel domain adaptation method for change detection. They modified the CycleGAN model to perform image translation over input image pairings to reduce the domain shift of various datasets.

In recent years, there have been significant advancements in cross-domain object detection of RS images. Several methods [16], [26], [29], [30], [31] have emerged, which are specifically designed for RS images. Koga et al. [31] have proposed the metric-based method with designed correlation alignment metric loss. There are also some adversarial-based methods [26], [29] proposed for cross-domain object detection in RS. For example, Chen et al. [29] have proposed Rotation-Invariant and Relation-Aware (RIRA)-Cross-Domain Adaptation Object Detection (CDAOD), which aligns the feature distribution at the prototype and image levels. Besides, Li et al. [30] have proposed the generative-based algorithm, which employs the CycleGAN and AgGAN to conduct image translation. Different from the methods mentioned above, Zhu et al. [16] have proposed RFA-Net to focus on the feature alignment and utilization of unlabeled target data. Several methods have demonstrated impressive performance in cross-domain object detection in RS. However, these techniques are still susceptible to the influence of biased information from the source domain. Consequently, pseudo-labels inevitably incorporate certain noisy predictions that pose challenges in terms of filtering them out. Therefore, we propose DualDA-Net to address this issue and make improvements to alleviate the effect of biased information, thus further improving the model's performance.

C. Mean Teacher and Co-Training

Mean teacher is first proposed by Tarvainen and Valpola [42], which ensembles model at temporal level with teacher–student framework. Many subsequent methods [34], [36], [41] have demonstrated the effectiveness of mean teacher in training with joint labeled and unlabeled data. For example, Liu et al. [34] adopt the mean teacher [42] paradigm to dynamically update pseudo-labels via a progressive teacher model. In addition, they use the strong–weak data augmentation to increase the detector's robustness when it is trained on noisy data. Nevertheless, the impact of noisy data on the model needs to be focused on. Therefore, co-training [56], [57] has been proposed, which is a representative paradigm of disagreement-based approaches, which generate multiple models and exploit the disagreement among them with unlabeled data. As for the cooperation of labeled and unlabeled data, co-training is frequently adopted in domain adaptation [58], [59], [60] and semisupervised

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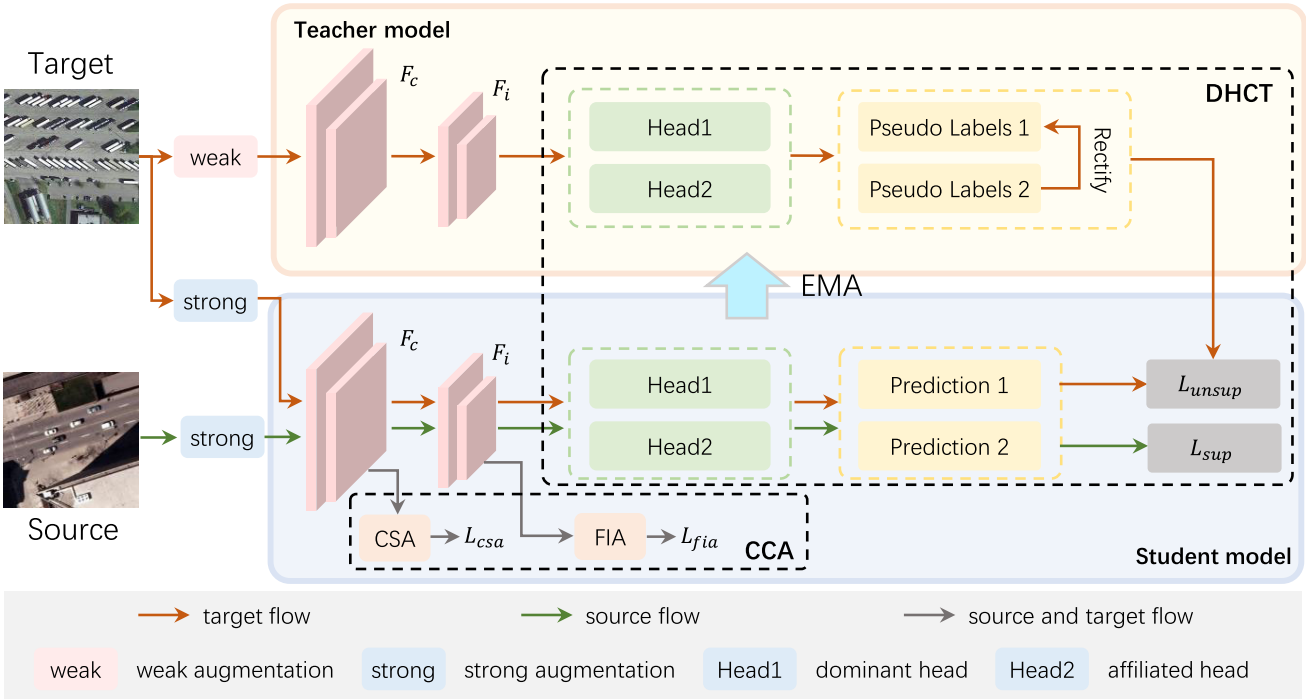


Fig. 3. Architecture of our Dual-Net. DualDA-Net is composed of CCA and DHCT. The CCA consists of CSA and FIA, which is deployed on multilevel feature alignment. DHCT benefits from the dual heads (dominant and affiliated heads), which predict the pseudo-labels with different thresholds. Moreover, the predictions of affiliated head rectify the predictions of dominant head for the pseudo-labels with high quality. Using an EMA, the teacher network is updated with student network parameters under the guidance of the teacher–student framework. Supported by the teacher–student framework, the teacher network is updated with student network parameters via EMA.

learning [41], [43], [44]. For example, the deep co-training with task decomposition (DeCoTa) method is suggested by Yang et al. [60] to divide the semisupervised domain adaptation task into two subtasks. The model with two heads is designed by the algorithm based on the well-known co-training framework for semisupervised learning and domain adaption, respectively. Chen et al. [41] propose an efficient algorithm multihead co-training (MHCT). In order to enable the use of unlabeled data and enhance the performance of the model, the MHCT incorporates base learners into a multihead structure. Inspired by co-training, we utilize multiple heads in the network to collaboratively enhance the benefits of mining unlabeled data while minimizing the training burden caused by multiple network structures. Specifically, we propose DualDA-Net for UDA object detection, which incorporates a DHCT strategy to correct noisy pseudo-labels. In addition, a teacher–student framework is employed for model training, where the teacher model predictions obtained from an ensemble of student models are used to generate pseudo-labels.

III. Methods

The proposed DualDA-Net is implemented in detail in this section, and the complete workflow of the DualDA-Net is shown in Fig. 3. DualDA-Net, supported by the teacher–student framework, consists of two core modules: CCA and DHCT.

A. Overview

1) Problem Definition: A labeled source domain $\mathcal{D}_s = \{X_s, Y_s\}$ and an unlabeled target domain $\mathcal{D}_t = \{X_t\}$ are provided for the algorithm. Y_s is the annotation of source image X_s, which is composed of the bounding boxes B_s and corresponding categories C_s. The target image X_t is unlabeled, and its annotation is unavailable. The cross-domain object detection technique is implemented utilizing $\mathcal{D}_s$ and $\mathcal{D}_t$. The detector $\mathcal{M}$ can ultimately be obtained utilizing such an algorithm and perform properly in the target domain $\mathcal{D}_t$.

2) Framework: As illustrated in Fig. 3, we propose DualDA-Net to improve performance on target domain data. The network is supported by the teacher–student framework, where the source images with annotations are only forwarded into the student network $\mathcal{M}^S$ with supervised loss. Pseudo-labels for unlabeled target data are predicted by the teacher network $\mathcal{M}^T$, treated as the student network's supervision. Concerning the performance of the detector, the unlabeled data with weak and strong data augmentation are fed into the teacher and student networks, respectively. Besides, the teacher network is progressively updated with student network parameters via exponential moving average (EMA). To address the impact of domain shift on cross-domain object detection, we propose employing a combination of CCA and DHCT. The CCA utilizes adversarial loss to align multilevel features with discriminators. The DHCT is applied to the detection head, where dual heads gradually correct each other. Specifically, one detection head acts as the affiliated head, progressively refining predictions for the other head (dominant head). By adopting this teacher–student framework with our proposed co-training approach, we enhance the detector's performance by utilizing more accurate pseudo-labels.

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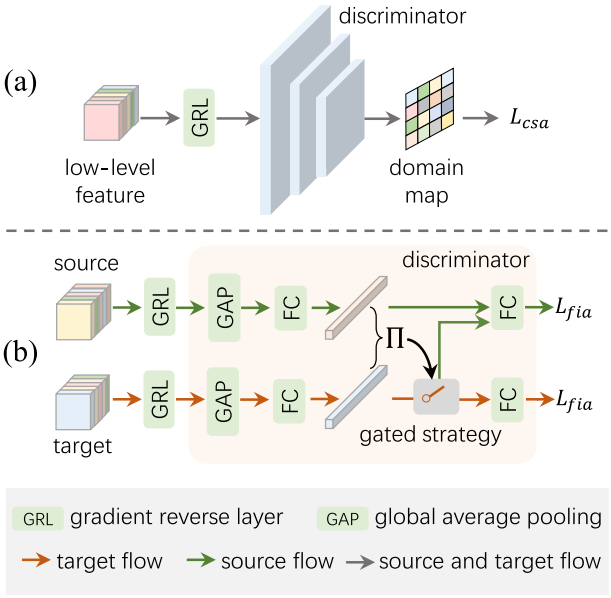


Fig. 4. Two components of CCA. (a) CSA injects the low-level features into the discriminator for predicting the domain probability map. (b) FIA designs domain-specific branches for source and target high-level features. Besides, a gated strategy is proposed to decide the route for the target domain features according to the deviation Π.

B. Proposed Method

The DualDA-Net is composed of the CCA and DHCT. In this section, we illustrate the specific implementation of CCA and DHCT.

1) Coarse-to-Fine Consistency Alignment: The domain shift between different RS datasets is mainly caused by variations in object position, angle, and image acquisition conditions. In RS object detection, the domain shift hinders the detector from extracting consistent feature distributions across different datasets, resulting in reduced accuracy in the target domain. Furthermore, due to the inconsistent feature distribution, the detector learns biased information from the source domain. Within the teacher–student framework, the teacher model fails to generate high-quality pseudo-labels for the unlabeled target data. To address this issue, we propose a CCA in the feature extraction phase of the pipeline. CCA aligns the feature distribution on multiple levels progressively, which is composed of coarse semantic alignment (CSA) and fine instance alignment (FIA). CSA is accomplished on low-level features with spatial adversarial loss, while FIA is achieved with a gated dual branch to align fine instance information. Specifically, we divide the backbone into the subnetworks F_c and F_i, as shown in Fig. 3. The features f_c and f_i generated by F_c and F_i are utilized for CSA and FIA, respectively.

a) Coarse semantic alignment: We propose a method for aligning the low-level features of different domains by designing a discriminator with adversarial loss. The low-level features capture coarse semantic information specific to their respective domains within limited receptive fields. To achieve coarse alignment, we use a subnetwork F_c to forward the input images x and obtain the low-level feature f_c. Then, a discriminator D_c with a GRL [28] is employed to predict domain labels d for every pixel location of the low-level feature f_c. The domain label d is assigned as 0 for the source domain and 1 for the target domain. The GRL is used to confuse the predictions of the domain discriminator D_c, ensuring that the feature distributions remain consistent. We use an adversarial loss $\mathcal{L}_{\text{csa}}$ to achieve this consistency, which is formulated as follows:

$$\mathcal{L}^s_{\text{csa}} = -\frac{1}{N_s HW}\sum_{i=1}^{N_s}\sum_{w=1}^{W}\sum_{h=1}^{H} D_c\left(F_c\left(x_i^s\right)\right)^2_{wh} \tag{1}$$

$$\mathcal{L}^t_{\text{csa}} = -\frac{1}{N_t HW}\sum_{i=1}^{N_t}\sum_{w=1}^{W}\sum_{h=1}^{H}\left(1 - D_c\left(F_c\left(x_i^t\right)\right)_{wh}\right)^2. \tag{2}$$

Equations (1) and (2) represent the adversarial loss for the source image x_i^s and target image x_i^t. Besides, N_s and N_t are the number of source and target images. The total adversarial loss of coarse alignment is summarized as follows:

$$\mathcal{L}_{\text{csa}}(F_c, D_c) = \frac{1}{2}\left(\mathcal{L}_c^s + \gamma\mathcal{L}_c^t\right). \tag{3}$$

The γ is the weight to balance the source and target adversarial losses.

b) Fine instance alignment: As the receptive field expands, higher-level features become enriched with more detailed semantic information about the object instance. We introduce a discriminator, D_f, incorporating a gated branch strategy to guarantee the consistency of feature distributions between the source and target domains. This strategy considers the correlation between the two domains adaptively. To extract these high-level features f_i, we use a subnetwork, F_i, which feeds into the global average pooling layer, $\mathcal{G}(\cdot)$. As shown in Fig. 4, the resulting channel information is forwarded into the respective branch using a fully connected layer. To reflect the information of feature distributions between cross-domain features, we design a domain statistics estimation, ψ, which is formulated as follows:

$$\psi_s = \frac{\mu\left(f^s_{\text{channel}}\right)}{\sqrt{\sigma\left(f^s_{\text{channel}}\right)+\epsilon}}, \quad \psi_t = \frac{\mu\left(f^t_{\text{channel}}\right)}{\sqrt{\sigma\left(f^t_{\text{channel}}\right)+\epsilon}} \tag{4}$$

in which the $\mu(\cdot)$ and $\sigma(\cdot)$ represent the calculation of mean and variance, respectively. ϵ is a constant, which is set as $1e-5$ to avoid invalid division. Besides, the f_{channel} is got as we describe above. The operations are formulated as follows:

$$\begin{cases} f^s_{\text{channel}} = \text{FC}_s\left(\mathcal{G}\left(f_i^s\right)\right) \\ f^t_{\text{channel}} = \text{FC}_t\left(\mathcal{G}\left(f_i^t\right)\right) \end{cases} \tag{5}$$

in which the $\text{FC}_s(\cdot)$ and $\text{FC}_t(\cdot)$ represent the fully connected layer in respective branch of source and target domains.

The domain statistics estimations ψ combine the mean and variance of the features in global channel. Subsequently, the gated branch strategy calculates the deviation Π between cross domains via domain statistical estimation and determines the way to perform prediction domain probability for target domain. The deviation Π is calculated by the formulation as follows:

$$\Pi = \tanh\left(\frac{\|\psi_s - \psi_t\|_p}{\psi_t}\right) \tag{6}$$

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in which the $\|\cdot\|_p$ is the p norm. We directly route the source domain features f^s_{channel} into their corresponding branches to predict domain probability p_s with each channel. However, for the target domain features f^t_{channel}, we control the branch where the target domain features route utilizing a threshold π. When the deviation Π between the cross domains is less than the threshold π, it means that the distribution between the source and target domains is similar. We route the target domain features f^t_{channel} to the branch of the source domain, sharing the parameters of the source domain. In contrast, we forward it into the target domain's own branch for domain probability p_t prediction. We formulate the gated strategy as follows:

$$p_s = \text{softmax}\left(\widetilde{\text{FC}}_s\left(f^s_{\text{channel}}\right)\right) \tag{7}$$

$$p_t = \begin{cases} \text{softmax}\left(\widetilde{\text{FC}}_s\left(f^t_{\text{channel}}\right)\right), & \text{if } \Pi < \pi \\ \text{softmax}\left(\widetilde{\text{FC}}_t\left(f^t_{\text{channel}}\right)\right), & \text{otherwise} \end{cases} \tag{8}$$

in which the $\widetilde{\text{FC}}_s$ and $\widetilde{\text{FC}}_t$ represent the fully connected layer in respective branch of source and target domains for probability prediction. Also, the threshold π is set as 0.2 in our implementation. Besides, we omit the activation function in (7) and (8) to simplify the representation. Once the probability predictions are obtained, the domain adversarial loss $\mathcal{L}_{\text{fia}}$ is calculated as follows:

$$\mathcal{L}^s_{\text{fia}} = -\frac{1}{N_s}\sum_{i=1}^{N_s}\text{focal}(p_s, 0) \tag{9}$$

$$\mathcal{L}^t_{\text{fia}} = -\frac{1}{N_t}\sum_{i=1}^{N_t}\text{focal}(p_t, 1) \tag{10}$$

in which the focal$(\cdot)$ means the focal loss function [61]. Finally, we summarize the loss function of discriminator D_f as follows:

$$\mathcal{L}_{\text{fia}}\left(F_i, D_f\right) = \frac{1}{2}\left(\mathcal{L}^s_{\text{fia}} + \beta\mathcal{L}^t_{\text{fia}}\right) \tag{11}$$

in which the parameter β is utilized to trade off the loss between cross domains.

We implement CCA through CSA and FIA and formulate the adversarial loss as follows:

$$\mathcal{L}_{\text{adv}} = \mathcal{L}_{\text{csa}}(F_c, D_c) + \mathcal{L}_{\text{fia}}\left(F_i, D_f\right). \tag{12}$$

2) Dual-Head Co-Training: The teacher–student framework comprises the teacher and student networks. To generate pseudo-labels, the teacher network applies weak augmentation to the unlabeled input. The student network, on the other hand, employs strong augmentation along with the pseudo-labels to compute the unsupervised detection loss. Regarding the weak–strong data augmentation, the strong augmentation acts as regularization for the student network's detection loss. Consequently, this weak–strong augmentation enhances the robustness of the model trained with noisy labels and facilitates knowledge distillation between the teacher and student networks. In DHCT, a two-stage object detector serves as the teacher and student networks. The RPN generates K_i object proposals for input image x_i, which are then sent to ROIHead for localization and classification using ROIAlign. The teacher network $\mathcal{M}^T$ predicts pseudo-labels for unlabeled data in the teacher–student framework. The predictions of ROIHead are filtered by a threshold τ to improve pseudo-label precision, and the filtered predictions serve as supervision for the unlabeled data. Although progressive alignment can align feature distributions across domains, biased domain information is learned in the source domain due to domain-specific characteristics, making it difficult to generalize to target domains and weakening the quality of pseudo-labels.

Specifically, some low-confidence true positives (TPs) are easily filtered out by the threshold τ and are difficult to be learned in the subsequent training process. Simultaneously, if the threshold τ is simply reduced, it tends to make the noisy labels difficult to be filtered. Therefore, we propose DHCT, in which the dual head contains a dominant head $\mathcal{R}_d$ and an affiliated head $\mathcal{R}_a$. The dominant head $\mathcal{R}_d$ is utilized to train unlabeled data with pseudo-labels generated with a high degree of confidence. Meanwhile, to lessen the impact of high-confidence false positives (FPs), the affiliated head $\mathcal{R}_a$, in the teacher network, corrects the pseudo-labels produced by dominant head $\mathcal{R}_d$.

We summarize our DCHT in Algorithm 1. As explained in Algorithm 1, for the labeled source domain data (x^i_s, y^i_s), we directly forward its features into the two heads to compute the detection loss, which is formulated as follows:

$$\mathcal{L}_{\text{sup}} = \sum_i^{N_s}\left(\mathcal{L}^{\mathcal{R}_d}_{\text{sup}}\left(x^i_s, y^i_s\right) + \mathcal{L}^{\mathcal{R}_a}_{\text{sup}}\left(x^i_s, y^i_s\right)\right) \tag{13}$$

in which $\mathcal{L}^{\mathcal{R}_d}_{\text{sup}}(\cdot)$ and $\mathcal{L}^{\mathcal{R}_a}_{\text{sup}}(\cdot)$ are the detection loss with regression and classification losses of $\mathcal{R}_d$ and $\mathcal{R}_a$, respectively. For unlabeled data in the target domain, the teacher network $\mathcal{T}$ is designed to predict pseudo-labels. Specifically, the DHCT framework generates pseudo-labels using dual-heads at this stage and refines the pseudo-labels of the dominant head using the pseudo-labels of the affiliated head. The predictions of $\mathcal{R}_d$ and $\mathcal{R}_a$ are formulated as follows:

$$\begin{cases} \left\{B^i_t, C^i_t\right\}_d = \mathcal{R}^S_d\left(\mathcal{B}^S\left(x^i_t\right)\right) \\ \left\{B^i_t, C^i_t\right\}_a = \mathcal{R}^S_a\left(\mathcal{B}^S\left(x^i_t\right)\right) \end{cases} \tag{14}$$

in which $\{B^i_t, C^i_t\}_d$ are the predicted bounding boxes and categories of x^i_t by $\mathcal{R}^S_d$ and $\{B^i_t, C^i_t\}_a$ are that by $\mathcal{R}^S_a$. Subsequently, these predictions are filtered by the corresponding thresholds τ^d and τ^a to obtain pseudo-labels. For the purposes of further rectification of the pseudo-labels, τ^d is greater than τ^a. The following is the formulation of the pseudo-labels produced by the two thresholds:

$$\begin{cases} \left\{B^i_t, C^i_t\right\}^{\text{pseudo}}_d = \left\{\left\{B^i_t, C^i_t\right\}^j_a \mid C^i_t > \tau_d\right\} \\ \left\{B^i_t, C^i_t\right\}^{\text{pseudo}}_a = \left\{\left\{B^i_t, C^i_t\right\}^j_a \mid C^i_t > \tau_a\right\}. \end{cases} \tag{15}$$

In particular, τ_d is greater than τ_a for the purpose of subsequent rectification of the pseudo-labels.

The dual-head rectification approach is applied to the pseudo-labels after they have been filtered by the relevant threshold to improve quality. Specifically, we elaborate such strategy as follows. Prior to doing anything more, we compute

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Algorithm 1 DHCT

Required: Teacher Network $\mathcal{M}^T = \{\mathcal{B}^T, \mathcal{R}_d^T, \mathcal{R}_a^T\}$, Student Network $\mathcal{M}^S = \{\mathcal{B}^S, \mathcal{R}_d^S, \mathcal{R}_a^S\}$, in which the $\mathcal{B}$, $\mathcal{R}_d$, $\mathcal{R}_a$ are the backbone, dominant head, and affiliated head of the networks M_T and M_S, respectively. Source domain dataset $\mathcal{D}_s = \{X_s, Y_s\}$ and target domain dataset $\mathcal{D}_t = \{X_t\}$, threshold τ^d for $\mathcal{R}_d^T$ and τ^a for $\mathcal{R}_a^T$ in $\mathcal{M}^T$.
Ensure: The threshold τ^d is larger than threshold τ^s.
for x_s^i *in* $\mathcal{D}_s$, x_t^i *in* $\mathcal{D}_t$ **do**
  Backbone $\mathcal{B}^S$ and two heads $\mathcal{R}_d$, $\mathcal{R}_a$ forward the source images x_s^i with strong augmentation to compute the detection loss;
  Backbone $\mathcal{B}^T$ of $\mathcal{M}^T$ forward the unlabeled target images x_t^i with weak augmentation to get the features;
  After generating K proposals for x_t^i, the proposals and features together are forwarded into the two heads $\mathcal{R}_d$, $\mathcal{R}_d$ to get the predictions,
  $\{B_t^i, C_t^i\}_d = \mathcal{R}_d^S\left(\mathcal{B}^S\left(x_t^i\right), Proposals\right)$,
  $\{B_t^i, C_t^i\}_a = \mathcal{R}_d^S\left(\mathcal{B}^S\left(x_t^i\right), Proposals\right)$;
  The thresholds τ^d and τ^a filter the predictions to get the pseudo labels,
  $\{B_t^i, C_t^i\}_d^{pseudo} = \{\{B_t^i, C_t^i\}_a^j \mid C_t^i > \tau_d\}$,
  $\{B_t^i, C_t^i\}_a^{pseudo} = \{\{B_t^i, C_t^i\}_a^j \mid C_t^i > \tau_a\}$;
  Calculate the IOU matrix to get the matched boxes,
  $Matrix_{IOU} = IOU\left(\{B_t^i\}_d^{pseudo}, \{B_t^i\}_a^{pseudo}\right)$
  $Box_{matched} = \{\{B_t^i, C_t^i\}_{a,j}^{pseudo} \mid Matrix_{IOU} > 0.9\}$;
  Compare the confidence score between $\{B_i^s, C_i^s\}_d^{pseudo}$ and $Box_{matched}$, and replace the boxes in $\{B_i^s, C_i^s\}_d^{pseudo}$ using the box with higher confidence score;
  $\{B_i^s, C_i^s\}_d^{pseudo} \leftarrow$ update by the matched boxes with higher confidence scores;
  Forward the target images after strong augmentation with pseudo labels into $\mathcal{M}^S$ for unsupervised detection loss;
  Loss backward and updating the parameters of networks.
end for

the intersection over union (IOU) between the pseudo-labels we acquired from the two heads

$$\text{Matrix}_{\text{IOU}} = \text{IOU}\left(\left\{B_t^i\right\}_d^{\text{pseudo}}, \left\{B_t^i\right\}_a^{\text{pseudo}}\right). \tag{16}$$

To better understand the rectifications, we visualize the process in Fig. 5. After that, we locate the matched boxes via the IOU matrix. In detail, we utilize the box pairs with an IOU greater than 0.9 as the matched boxes

$$\text{Box}_{\text{matched}} = \left\{\left\{B_t^i, C_t^i\right\}_{a,j}^{\text{pseudo}} \;\middle|\; \text{Matrix}_{\text{IOU}} > 0.9\right\}. \tag{17}$$

In relation to the matched pairs of bounding boxes, we compare the confidence scores between the boxes of the dominant head and the affiliated head. Ultimately, we consider the box with the higher confidence score as the pseudo-label for that particular position. We rectify the pseudo-labels generated by

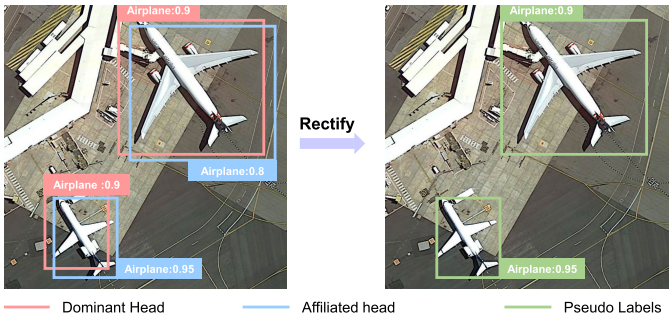


Fig. 5. Pseudo-labels from the dominant head are shown in red rectangles, and those from the affiliated head are shown in blue rectangles. We match the pseudo-labels of different heads via computing the IOU between them. After that, for the matched pseudo-labels in the same location, we utilize the pseudo-label with a higher confidence as the final pseudo-label, which is represented by the green rectangles.

$\mathcal{R}_d^T$ with pseudo-labels generated by $\mathcal{R}_a^T$ as follows:

$$\left\{\left\{B_t^i, C_t^i\right\}_d^j\right. = \begin{cases} \left\{\left\{B_t^i, C_t^i\right\}_a^j, & \text{if matched and } \left\{C_t^i\right\}_d < \left\{C_t^i\right\}_a \\ \left\{\left\{B_t^i, C_t^i\right\}_d^j, & \text{otherwise.} \end{cases} \tag{18}$$

After acquiring the rectified pseudo-labels for unlabeled target data, we forward the target image with corresponding pseudo-labels into the student network $\mathcal{M}^S$ and compute the detection loss. The loss treated as unsupervised loss is calculated as follows:

$$\mathcal{L}_{\text{unsup}} = \sum_i^{N_t}\left(\mathcal{L}_{\text{unsup}}^{\mathcal{R}_d}\left(x_t^i, y_t^{d,i}\right) + \mathcal{L}_{\text{unsup}}^{\mathcal{R}_a}\left(x_t^i, y_t^{a,i}\right)\right) \tag{19}$$

in which $y_t^{d,i}$ and $y_t^{a,i}$ are the pseudo-labels generated by dominant head and affiliated head, respectively.

C. Model Initialization and Update

At first, we initialize the student network $\mathcal{M}^S$ via the pretrained model. To ensure that two different heads in the student model converge independently during training, we utilize random parameter initialization for detection heads. In addition, different data augmentation methods are used for the input data of the same mini-batch. We also propose a wake-up stage, for the sake of initializing the teacher network $\mathcal{M}^T$ better and generating high-quality pseudo-labels. In the wake-up phase, the student network $\mathcal{M}^S$ forwards source and target domain data and is updated by backpropagation utilizing the loss in (3), (11), and (13). At the end of the wake-up phase, we copy the parameters of the student network $\mathcal{M}^S$ to initialize the teacher network $\mathcal{M}^T$. In the subsequent training process, the teacher network $\mathcal{M}^T$ makes predictions on the unlabeled target data, while its parameters are updated by the student network $\mathcal{M}^S$ with the exponential sliding average. For a given number of update iterations, the specific update is summarized as follows:

$$\theta_T^i \Leftarrow \alpha\theta_T^i + (1-\alpha)\theta_S^i \tag{20}$$

in which the θ_T^i and θ_S^i are the parameters of $\mathcal{M}^T$ and $\mathcal{M}^S$, respectively. α is the update rate.

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D. Overall Objective

With the consideration of the losses in CCA as well as DHCT, we summarize the total loss and formulate it as follows. Concerning the adversarial learning loss in feature extraction phase, we summarize the adversarial loss as follows:

$$\mathcal{L}_{\text{adv}} = \mathcal{L}_c(F_c, D_c) + \mathcal{L}_f\left(F_i, D_f\right) \tag{21}$$

in which $\mathcal{L}_c(\cdot)$ and $\mathcal{L}_f(\cdot)$ are the adversarial loss in (3) and (11), respectively. Besides, we also summarize the detection loss, which is composed of the supervised loss and unsupervised loss and is formulated as follows:

$$\mathcal{L}_{\text{det}} = \mathcal{L}_{\text{sup}} + \xi\mathcal{L}_{\text{unsup}} \tag{22}$$

in which the $\mathcal{L}_{\text{sup}}$ and $\mathcal{L}_{\text{unsup}}$ are the loss in (13) and (19), respectively. The ξ is the hyperparameter to trade off the supervised and unsupervised losses. The total loss is formulated as follows:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{det}} - \eta\mathcal{L}_{\text{adv}} \tag{23}$$

in which $\mathcal{L}_{\text{adv}}$ is the adversarial loss in (12) and η is the parameter to balance the detection loss and adversarial loss.

IV. Experiments and Results

A. Datasets

1) NWPU VHR-10 Dataset [12]: The dataset containing 800 images is made up of 85 pansharpened color infrared images from the Vaihingen dataset [65] and 715 color Google Earth images. The spatial resolution of the images from Google Earth ranges from 0.5 to 2 m, while that from the Vaihingen dataset is 0.08 m. The ten classes of NWPU VHR-10 are split into sets representing both positive and negative images. At least one object is visible in each of the 650 positive images. In addition, there are no objects in the 150 image negative image collection. With a ratio of 14:6:5, the images in the positive image set are split into the train, validation, and test sets.

2) DOTA Dataset [13]: The DOTA dataset includes 2806 aerial images from various sensors and platforms (such as Google Earth). The photographs have a dimension of roughly 4000 × 4000 pixels. The experts have annotated a total of 188 282 items with a wide range of scales, orientations, and forms covering 15 common object categories. We choose the cases in DOTA that fall into the same categories in the NWPU VHR-10 dataset to rebuild the dataset in order to more effectively compare the performance of the cross-domain object detection techniques. The DOTA dataset is treated using the same ten categories as the DOTA* dataset.

3) WHU Buildings Change Detection Dataset [66]: This dataset consists of 12 796 buildings in 20.5 km^2 from aerial images captured in April 2012 (16 077 buildings in the same area in 2016 dataset). The aerial image has a 0.2-m resolution. The region has a earthquake in February 2011 and was rebuilt in the subsequent years. The temporal offset causes a domain offset between the two images. We regard the images in 2012 (WHU 2012) as the source dataset and the other in 2016 (WHU 2016) as target domain. The images are cropped into 512 × 512 patches. We generate the bounding boxes for the building according to the masks as the experimental settings described by Chen et al. [29].

4) RarePlanes Dataset [67]: The RarePlanes [67] is the largest publicly accessible dataset comprising actual and synthetic images combined at very high quality. The simulation platform of Reverie uses AI to produce the synthetic imagery dataset. It includes 630 000 annotations from aircraft on 50 000 fake satellite photographs that simulate a total area of 9331.2 km^2. Also, the synthetic image dataset covers 15 regions. For the purpose of conducting the synthetic to real domain adaptation, we choose at random one region containing the synthetic images (the region of Miami is selected by a randomly generated random number within [1, 15]). We formulate the images as the RarePlanes-Miami dataset, and the category of instances is treated as the airplane without the fine-grained category.

5) FAIR1M Dataset [2]: A unique benchmark dataset with more than one million instances is the FAIR1M dataset [2]. More than 15 000 images with fine-grained instances are annotated by oriented bounding boxes with regard to five categories and 37 subcategories. We extract the airplane in the dataset to formulate the FAIR-Airplane for the synthetic to real domain adaptation.

B. Implementation Details

Based on the teacher–student framework, our DualDA-Net utilizes a two-stage object detector, Faster-RCNN [46], to implement domain adaptation object detection. In our experiments, ResNet [68] and VGG [69] are utilized as our backbone. The pretrained model on ImageNet initializes the backbone. As for ResNet, the features output from Stage2 and Stage4 output is used for CSA and FIA, respectively. In addition, VGG, we use the features output from Stage2 and Stage5 of VGG for CCA. We develop the model until convergence and then infer it on the test dataset. Moreover, we set the iteration of wake-up stage as 6000 for the initialization of teacher network. In both the source data and the target data, we employ a learning rate of 0.005 and a batch size of 2. In addition, after 10 000 iterations, the learning rate is decreased by ten times. The given decay rates for momentum and weight are 0.9 and 0.0001, respectively. We utilize the PyTorch framework to create our DualDA-Net. On the 24-GB NVIDIA GeForce RTX 3090 GPU, all of our experiments are performed.

For the teacher–student framework, we followed the experimental settings in Li et al. [70]. Our weak data augmentation involves random horizontal or vertical flips, while our strong data augmentation includes color jittering, gray scale, Gaussian blur, and cutout [71]. We update the teacher network using the student network through EMA every iteration with an update rate of α set to 0.9996 for progressive update. In addition, we set the parameters ξ and η in (22) and (23) to 3.5 and 2.5, respectively. For our CCA implementation, we adopt γ and β in (3) and (11) as 1 to balance the source and target adversarial losses. The p norm in (6) is set to the absolute-value norm with p equal to 1. In our DHCT approach, we set the thresholds τ^d and τ^a for the dominant and affiliated

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TABLE I
COMPARATIVE ANALYSIS OF THE NWPU VHR-10 TO DOTA EXPERIMENTS WITH THE SOURCE ONLY AND VARIOUS DOMAIN ADAPTION OBJECT DETECTION TECHNIQUES. THE BEST AP AND MAP VALUES ARE THOSE IN BOLD

Method	Backbone	Airplane	Ship	Storage tank	Baseball diamond	Tennis court	Basketball court	Ground track field	Harbor	Bridge	Vehicle	mAP(%)
Source-only	VGG16	48.21	7.02	14.83	59.55	58.39	36.55	27.15	10.25	1.09	9.09	27.31
SWDA [62]		49.05	9.09	18.4	56.46	67.33	41.05	32.89	11.74	0.31	10.69	29.70
HTCN [63]		53.06	8.97	17.33	56.66	59.8	56.46	26.87	13.33	0.83	9.09	30.24
SCL [64]		45.52	9.39	22.05	55.45	64.92	49.29	29.41	13.42	0.87	7.58	29.79
RFA-Net [16]		52.95	9.07	15.12	60.93	67.89	44.37	31.37	10.04	2.75	10.4	30.49
Ours(DualDA-Net)	VGG16	61.13	0.25	25.81	53.97	87.38	65.52	27.92	7.19	0.28	0.63	**33.01**
Source-only	Res101	42.82	9.09	8.26	42.14	53.30	29.85	24.19	9.43	1.07	6.82	22.70
SWDA [62]		55.32	10.09	16.04	65.03	70.03	31.17	43.77	15.00	9.09	11.08	32.66
HTCN [63]		54.27	10.32	13.83	60.70	62.99	33.81	32.17	10.92	9.10	11.74	29.99
SCL [64]		52.05	10.90	16.54	59.88	68.69	33.67	38.55	10.72	5.20	10.82	30.70
RFA-Net [16]		52.31	9.79	21.36	65.86	78.38	39.80	41.94	11.48	1.82	11.01	33.30
Ours(DualDA-Net)	Res101	59.67	1.46	28.24	65.74	88.41	85.63	39.45	8.53	0.40	0.79	**37.83**
Ours(DualDA-Net)	Res101+FPN	62.67	2.14	28.66	62.58	89.05	88.16	51.65	8.63	0.42	1.02	**39.50**

heads to 0.9 and 0.8, respectively. Notably, the final model result is generated by the main head of the teacher network.

The average precision (AP) is the measure we apply to assess the performance of the detector. The average of precision under several recalls is used to determine the AP. The formulation for the precision and recall is as follows:

$$\text{Precision} = \frac{N_{\text{TP}}}{N_{\text{TP}} + N_{\text{FP}}} \tag{24}$$

$$\text{Recall} = \frac{N_{\text{TP}}}{N_{\text{TP}} + N_{\text{FN}}} \tag{25}$$

in which the N_{TP}, N_{FP}, and N_{FN} represent the amount of the true positives (TPs), false positives (FPs), and false negatives (FNs), respectively. In particular, TP is the occurrence in the data that is identified and accurately categorized. False instance that is incorrectly discovered or classified is known as FP. Besides, the FN is the instance that is not localized. When the prediction's IOU from the ground truth is more than 0.5, it can be considered the true prediction (TP) during the inference phrase. The prediction appears under the FP category if not. In addition, the AP is determined by averaging the precision for a group of recall rates S that are distributed evenly. In our evaluation, we utilize $S = \{0, 0.1, \ldots, 1\}$, and the AP is defined as follows:

$$\text{AP} = \frac{1}{11} \sum_{r \in S} \text{Precision}|_{\text{Recall}=r}. \tag{26}$$

Also, the mAP is the mean of the total classes, which is treated as the evaluation in our experiments for object detection.

C. Experimental Results and Analyses

We undertake experiments in three different experimental settings—cross-dataset adaptation, cross-time adaptation, and synthetic to real adaptation—in order to properly assess the efficiency of the domain adaptation object detection methods in RS. The specific experimental settings and experimental results are presented as follows.

1) Cross-Dataset Adaptation: Cross-dataset adaptation involves different datasets collected with various satellite sensors in different regions. The variation and diversity in objects and backgrounds cause domain shift between the cross datasets. In this study, we use the DOTA dataset as the target dataset and the NWPU VHR-10 dataset as the source dataset. We organize the DOTA validation set to assess the effectiveness of the domain adaptation methods. Experimental results from the cross-dataset adaptation from the NWPU VHR-10 to DOTA datasets are shown in Table I.

Source only in Table I denotes the result of the model only being trained with source domain data. Comparing with the results of source only, our DualDA-Net improves 5.70 and 15.05 AP under the backbone of VGG16 and Res101, respectively. It demonstrates our algorithm's success in achieving a huge improvement on the target domain via the cooperation of CCA and DHCT. We also contrast our approach with the recent domain adaptation object detection methods. The outcomes of the experiments show how superior our algorithm is. Among these techniques, feature alignment is widely used by strong-weak distribution alignment (SWDA), Hierarchical Transferability Calibration Network (HTCN), and stacked complementary losses (SCL) to reduce the domain shift. However, the potential learnable knowledge in the target domain is ignored by SWDA, HTCN, and SCL, which leads to low performance. In addition, RFA-Net generates pseudo-labels through the model developed in the source domain. Such techniques frequently result in biased information in the pseudo-labels. DualDA-Net takes into account the domain shift between cross domains in feature-level alignment. Besides, the effects of biased information in the source domain are also alleviated by the CSA and DHCT. As shown in Fig. 6, it is obvious that the domain shift between the source and destination domains is significantly alleviated by our DuaDA-Net. As a result, DualDA-Net can effectively employ the teacher–student framework to transfer knowledge from the source domain into the target domain based on a consistent distribution. We also visualize some detection results and the feature map in Fig. 7 to demonstrate the effectiveness of our DualDA-Net. The detection results demonstrate that our method exhibits commendable performance in densely populated environments, particularly in the airplanes and storage tank categories. Moving forward, we acknowledge that further refinement is necessary for our proposed DualDA-Net to achieve optimal results in some specific categories.

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(a) Source-Only [46] (b) SWDA [62] (c) HTCN [63]

(d) SCL [64] (e) RFA-Net [16] (f) DualDA-Net

Fig. 6. Visualization of feature distributions between source and target domains via t-SNE [11]. We visualize the feature distributions between source and target domains after feature alignment. (a) Feature distribution without adaptation. (b)–(e) Feature distribution after adaptation via some comparable algorithms [16], [62], [63], [64]. (f) Feature distribution after adaptation by our proposed method; it can be observed that our feature embedding distribution is consistently much better than the previous approaches.

2) Cross-Time Adaptation: In cross-time adaptation, we conduct experiments using satellite image datasets from the same region with temporal offset. Aerial imageries taken at several periods in the same region will invariably show domain shifts due to changes in the buildings and plants. In the WHU buildings change detection [66] dataset, we serve the WHU 2012 dataset as the source domain and the WHU 2016 as the target domain. We utilize DualDA-Net for building detection and compare its performance with several existing domain adaptation algorithms. The backbone of the experiments is Res50 following the experimental settings as Chen et al. [29]. As shown in Table II, image level indicates the model conducts alignment via adversarial loss in the feature extraction stage, ins level indicates alignment on instance features extracted by ROIAlign via adversarial loss, and pseudo-label indicates that the model forwards the unlabeled data with pseudo-label.

TABLE II

CROSS-TIME ADAPTATION SCENARIO MEAN AP, WHU 2012 DATASET ⟶ WHU 2016 DATASET

Method	Img-level	Ins-level	Pseudo label	House AP
Source-only				71.99
CycleGAN [72]	✓(GAN-based)			70.11
AgGAN [73]	✓(GAN-based)			72.65
DA-Faster* [74]	✓			73.45
DA-Faster [74]	✓	✓		75.00
Chen et al. [29]	✓			77.09
SWDA [62]	✓			79.61
HTCN [63]	✓			80.33
SCL [64]	✓			80.08
RFA-Net [16]	✓		✓	80.23
DualDA-Net	✓		✓	**91.52**
Oracle				81.43

As demonstrated in Table II, imageries in the same region at different periods bring out a domain shift due to factors, such as building reconstruction. The detection models trained by the previous data have difficulty in generalizing on the new period. Although the GAN-based strategy is intended to use style transfer to align the distribution discrepancies between the source and target domains, the variety of situations and backgrounds makes it challenging to get better results with such approach. Through feature alignment, these domain adaptation techniques are successful in enhancing performance in the target domain. Compared with GAN-based techniques, they effectively eliminate the domain shift. In particular, compared with the model (Oracle) trained on the target domain with annotations, our strategy not only surpasses the other methods but also boosts performance by around 10 AP. Because of this, we may employ the target domain's pseudo-labels to fully exploit the potential of unlabeled data, enhancing the model's performance. The qualitative results of the detection are shown in Fig. 7.

3) Synthetic to Real Adaptation: With the development of virtual synthetic techniques, creating 3-D environments and instances has become much easier. This technique is very useful in several fields, as it can generate datasets along with corresponding data annotations, leading to the development of synthetic datasets. However, even when

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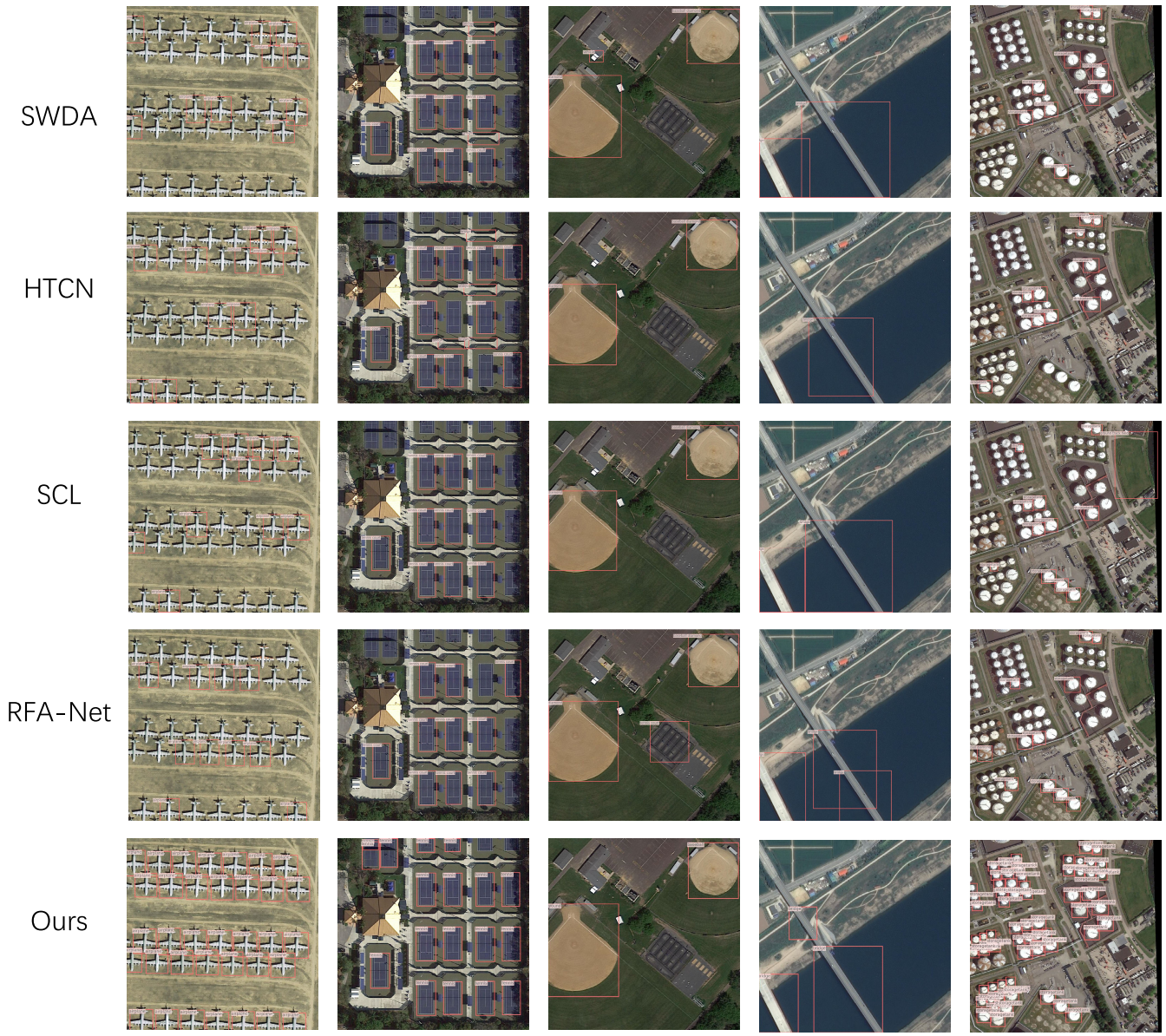


Fig. 7. From left to right, they are the qualitative detection results of cross-dataset adaptation. We visualize the detection results of SWDA [62], HTCN [63], SCL [64], RFA-Net [16], and our DualDA-Net.

models are trained on these synthetic datasets, they tend to perform poorly in real scenarios. To evaluate the effectiveness of domain adaptation models, we conducted an experiment using the synthetic dataset RarePlanes-Miami [67] as the source domain and the fine-grained aircraft dataset FAIR-Airplane [67] as the target domain with the same parameter settings as cross-dataset adaptation. In particular, we used 0.8 and 0.6 as the dominant head and affiliated head thresholds, respectively. Performance under the experimental conditions of synthetic to real adaptation is shown in Fig. 8.

Fig. 8 reveals that the model trained on synthetic data does not perform well on real data, demonstrating the presence of domain shift between the synthetic dataset and real data. To better utilize synthetic data and reduce annotation burden on real data, some domain adaptation object detection methods have been proposed to improve model generalization from source and to domains. However, as seen in Fig. 8, certain domain adaptation methods do not perform well on this experiment. For instance, SCL [64] and HTCN [63] rely on feature alignment between source and target domains at various levels but fail to yield satisfactory performance due to differences in instances between the two domains. RFA-Net, which applies a self-training approach to generate pseudo-labels for unlabeled target data, also suffers from biased information from the source domain affecting the quality of pseudo-labels and subsequent training. DualDA-Net proposes a CCA for aligning feature distributions at multiple levels where a gated branch strategy progressively aligns high-level features in the source and target domains. Furthermore, DHCT is used for pseudo-label co-rectification based on a teacher–student framework. The collaboration between CCA and DHCT enables

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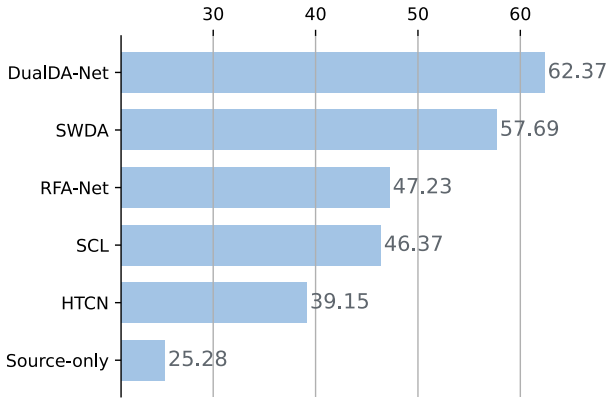


Fig. 8. Histogram of the experimental results in synthetic to real adaptation. From top to bottom, we display the results in descending order.

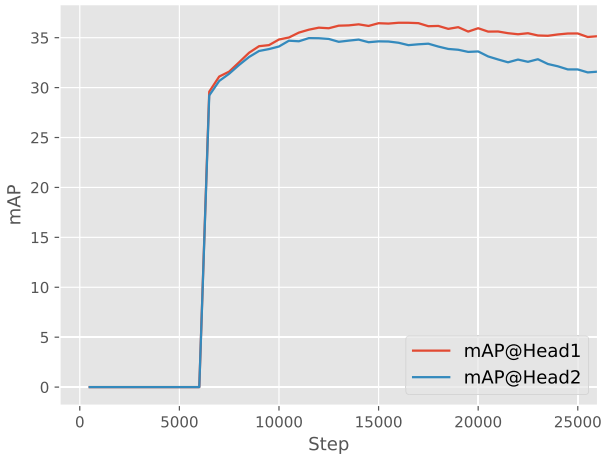


Fig. 9. Efficiency of training process on cross-dataset adaptation. During training process, the mAP of two heads of teacher model is displayed.

DualDA-Net to achieve excellent performance in synthetic to real adaptation experiments.

Drawing upon the results from the experiment on synthetic to real adaptation, we conclude that synthetic RS imagery plays a crucial role in RS object detection. Synthetic imagery offers the advantage of automated annotation generation and is more accessible compared with real RS data. By employing domain adaptation object detection algorithms, the domain shift between synthetic and real datasets can be minimized, allowing models trained on synthetic data to exhibit strong performance in real-world scenarios. Moreover, this approach alleviates the challenges associated with annotating real-world datasets.

4) Analysis of Training Efficiency: We evaluate the training efficiency of our proposed DualDA-Net by visualizing the mAP results of DOTA dataset during the training process in Fig. 9. Our experiments are conducted on a single 3090 GPU using a batch size of 2. We train the model for cross-dataset adaptation with a total of 28 000 iterations, out of which the first 6000 iterations are used for wake-up training. It takes approximately 8 h to complete all the training. After the wake-up stage, the teacher network is initialized with the student network, enabling the teacher network to have certain generalization on the test set. Subsequently, teacher network generates high-quality pseudo-labels for the unlabeled training set in the target domain. Based on the wake-up stage, the dominant head and affiliated head could quickly converge after 10 000 iterations. Therefore, we can conclude that DualDA-Net, based on the teacher–student framework with a wake-up stage, can achieve satisfactory results with fast convergence training efficiency.

D. Ablation Study

In this section, we conduct an ablation study on each module of DualDA-Net and analyze the threshold's impact on cross-dataset adaptation. The study evaluates the effectiveness of the modules and examines the threshold for the dominant head and affiliated head in cross-dataset adaptation. Furthermore, we ablate some other hyperparameters under cross-time adaptation, for example, the unsupervised weight ξ of loss $\mathcal{L}_{\text{det}}$ in (22), the adversarial weight η of loss $\mathcal{L}_{\text{total}}$ in (23), and the update rate α in (20).

TABLE III
ABLATION STUDY OF THE COMPONENTS IN OUR PROPOSED DUALDA-NET

	Backbone	CSA	FIA	DHCT	mAP
w/ CSA	ResNet101	✓			35.32
w/ FIA			✓		35.51
w/ DHCT				✓	35.76
w/o DHCT		✓	✓		35.58
w/o FIA		✓		✓	35.96
w/o CSA			✓	✓	36.40
DualDA-Net		✓	✓	✓	**37.83**

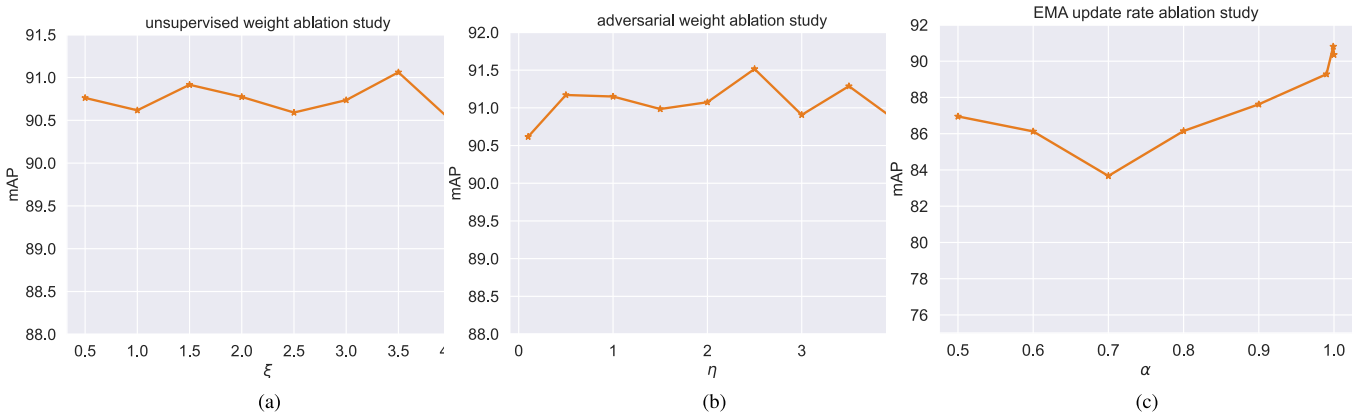
1) Analysis of Each Module: To demonstrate the effectiveness of our proposed DualDA-Net, we conduct an ablation study. First, we assess the performance of each individual component by planting it separately. Then, we combine arbitrary two components to validate their relationship. For the DCHT component, which requires two heads (dominant and affiliated), we ensured that both are present. In cases where DHCT is not available, we conduct experiments using a threshold value of 0.8 for a single head. All experiments are conducted using the ResNet101 backbone. The results of the ablation study are presented in Table III, where "w/" and "w/o" indicate whether the experiment is conducted with or without a specific module.

Benefiting from the utilization of unlabeled target data by the teacher–student framework, we can observe that when only a single module is utilized, DualDA-Net still outperforms the other baselines compared with the other algorithms shown in Table I. Regarding DualDA-Net, there is minimal distinction when employing a single module. Models equipped with a single CSA or FIA exhibit inferior performance compared to those with a single DHCT. However, by aligning the features in both the source and target domains gradually, from coarse to fine, the incorporation of CSA and FIA in the model mitigates the impact of domain shift. The model with CSA and FIA improves by 0.44 and 0.25 AP, respectively, as compared with the single CSA and FIA. However, the model's performance is diminished by 0.18 AP as a result of DHCT's absence of pseudo-label rectification. When the CSA or FIA is with the

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TABLE IV
ABLATION STUDY OF THE THRESHOLD OF DOMINANT HEAD AND AFFILIATED HEAD IN DUALDA-NET

Backbone	Head1	Head2	Airplane	Ship	Storage tank	Baseball diamond	Tennis court	Basketball court	Ground track field	Harbor	Bridge	Vehicle	AP
Res101	0.8	0.5	60.84	2.11	29.15	49.56	87.38	84.90	41.13	6.90	0.07	2.08	36.41
		0.6	60.56	2.29	29.22	57.71	87.70	84.54	37.39	7.51	0.11	2.00	36.90
		0.7	60.65	2.04	29.94	53.33	87.68	84.33	37.52	6.69	0.15	2.32	36.47
		0.8	61.52	2.43	30.01	61.04	88.24	76.56	36.76	5.95	0.14	1.16	36.38
	0.9	0.5	59.74	1.67	27.72	55.07	89.63	85.42	37.96	5.93	0.16	1.63	36.49
		0.6	60.19	2.20	29.88	55.25	89.36	83.26	40.39	7.25	0.17	1.05	36.90
		0.7	60.32	1.47	28.57	54.12	88.89	87.88	38.15	7.90	0.56	0.66	36.85
		0.8	59.67	1.46	28.24	65.74	88.41	85.63	39.45	8.53	0.40	0.79	**37.83**
Res101+FPN	0.9	0.6	61.40	2.46	34.36	41.98	86.24	88.14	46.99	6.36	0.23	5.68	37.38
		0.8	62.67	2.14	28.66	62.58	89.05	88.16	51.65	8.63	0.42	1.02	**39.50**



(a) (b) (c)

Fig. 10. Cross-time adaptation ablation study with various hyperparameters. (a) Experimental results with different unsupervised weights ξ. (b) Experimental results with different adversarial weights η. (c) Experimental results with different EMA update rates α.

DHCT, the model improves more AP with the assistance of the DHCT.

We draw the conclusion from the ablation investigation that all DualDA-Net components are essential for the cross-domain object detection of RS. The domain shift between the cross domains is mitigated by the coarse-to-fine feature alignment. In order to enhance the quality of the pseudo-labels for the dominant head, the DHCT also performs pseudo-label rectification. The model with the CSA, FIA, and DHCT finally turns in a commendable performance.

2) Analysis of Threshold in Different Heads: It is generally accepted that the quality of pseudo-labels increases with higher threshold values. However, when the threshold value of the affiliated head is fixed, the dominating head's threshold value does not necessarily correlate with performance. According to Table IV, when the affiliated head's threshold value is 0.6, the dominating head's threshold values of 0.9 and 0.8 produce the same performance. When the threshold is 0.9 instead of 0.8, the results are not better for each individual category. We provide the following explanation for this situation.

With respect to this, we explain as follows. Although high thresholds yield pseudo-labels with high quality, some predictions with low classification scores are also filtered by the thresholds, leading to the occurrence of some FNs. Meanwhile, low thresholds in the affiliated head can cause the occurrence of some FPs. The FN and FP affect the performance of the dominant head during the training process. Therefore, there is a certain trade-off between the dominant head and the affiliated head for the maximum benefit of the model. In the experiments of cross-dataset adaptation, we utilize 0.9 and 0.8 as the threshold for the dominant and affiliated heads, respectively. As shown in Table IV, the model achieves the best performance in both the experiments with Res101 as backbone and Res101 with FPN.

3) Analysis of Some Other Hyperparameters: To explore the effect of more refined hyperparameter values on model performance, we analyze three different hyperparameters separately and show the experimental results in Fig. 10.

Specifically, as for the unsupervised weight ξ of loss $\mathcal{L}_{\text{det}}$ in (22), it balances the supervised and unsupervised losses. As shown in Fig. 10(a), various experimental results are produced by different ξ values, and the best performance are qualified when ξ is 3.5. However, when ξ varies from 0.5 to 4.0, the uncertainty of the experimental results does not exceed 0.5 mAP. The ablation study leads us to the conclusion that the model performs best when ξ is 3.5, and comparable results can be obtained when ξ varies. We also explore the influence of the different η values. As shown in Fig. 10(b), the model perform bests when η is 2.5. When η is small, the feature alignment loss cannot be well optimized. Furthermore, a large η causes feature alignment overfitting, which degrades detection performance.

We have conducted experiments to investigate the effect of EMA update rate on model performance in the teacher–student

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framework. To accomplish this, we utilize different values of α ranging from 0.5 to 0.9 with an interval of 0.1. In addition, we have tested 0.99, 0.999, and 0.9999 to obtain refined parameters. As depicted in Fig. 10(c), the model performs poorly when α is between 0.5 and 0.9, exhibiting a large disparity in performance. However, as α approaches 1, the model achieved more satisfactory results. This is because a small EMA update rate influences the model's robustness, since the student model's parameters have been weighted more in each update. It potentially lessens the impact of the student model trained on target data with noisy labels. We have also observed that the model's performance decreases when α varies from 0.999 to 0.9999, indicating that the teacher model is updated slowly, resulting in slower model performance growth. Therefore, we set α as 0.9996 in all of our experiments.

V. Limitations and Future Work

In cross-time adaptation and synthetic to real adaptation, our proposed DualDA-Net achieves superior performance on a single class of instances (house or airplane). However, in Table I, DualDA-Net fails in some categories (ship, bridge, and vehicle) with special sizes and aspect ratios. For domain adaptation object detection, the domain gap between the source and target domains is not only reflected by the discrepancy in the distribution of background and instances at pixel level. In our future study, we aim to focus on specific instance features using oriented object detection, which can reduce the influence of the background surrounding the instance. Another element influencing performance is the variation in category distribution between the source and target domains. It is important to address the differences in intraclass and interclass distributions. By incorporating domain-specific features, such as color or texture, we can enhance the model's ability to distinguish between similar classes in the target domain. Moreover, the noisy labels in predictions cannot be removed completely. Label smoothing is a technique for reducing the impact of noisy labels. Besides, curriculum learning can also be used to reduce the noise in the pseudo-labels. The model is trained on easier examples first and gradually introduced to more difficult examples, which can help improve the performance.

VI. Conclusion

In this research, we suggest using the DualDA-Net for cross-domain object detection in RS imagery. Our proposed DualDA-Net is based on the teacher–student framework and consists of CCA and DHCT. The CCA is composed of CSA and FIA, which are, respectively, deployed on different feature levels for feature alignment. The CSA is conducted on the low-level feature with a spatial adversarial loss, while the FIA aligns the fine instance information via a gated dual branch. In addition, the DHCT deploys a co-training technique on the detection head with dual heads for the teacher and student networks, which produces high-quality pseudo-labels for unlabeled target data. By conducting experiments on cross-dataset adaptation, cross-time adaptation, and synthetic to real adaptation, we have shown the efficiency of our DualDA-Net. We also perform the ablation study on the components in our DualDA-Net, and the experimental results illustrate the contribution of each of the components and the association between them. Finally, we investigate the limitation of DualDA-Net in performing well in all categories. To enhance the model's performance across all categories, we will involve oriented object detection, domain-specific alignment, and curriculum learning in improving the performance. We anticipate that our work will inspire other cross-domain object detection research in RS imagery.

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Yangguang Zhu received the B.Sc. degree from the University of Science and Technology of China, Hefei, China, in 2019. He is currently pursuing the Ph.D. degree with the University of Chinese Academy of Sciences, Beijing, China, and the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing.

His research interests include computer vision and remote sensing interpretation, especially on object detection domain adaptation.



Xian Sun (Senior Member, IEEE) received the B.Sc. degree from the Beijing University of Aeronautics and Astronautics, Beijing, China, in 2004, and the M.Sc. and Ph.D. degrees from the Institute of Electronics, Chinese Academy of Sciences, Beijing, in 2009.

He is currently a Professor with the Aerospace Information Research Institute, Chinese Academy of Sciences. His research interests include computer vision, geospatial data mining, and remote sensing image understanding.



Wenhui Diao (Member, IEEE) received the B.Sc. degree from Xidian University, Xi'an, China, in 2011, and the M.Sc. and Ph.D. degrees from the Institute of Electronics, Chinese Academy of Sciences, Beijing, China, in 2016.

He is currently an Associate Professor with the Aerospace Information Research Institute, Chinese Academy of Sciences. His research interests include computer vision and remote sensing image analysis.



Haoran Wei (Graduate Student Member, IEEE) received the B.Sc. degree from the Minzu University of China, Beijing, China, in 2018. He is currently pursuing the Ph.D. degree with the University of Chinese Academy of Sciences, Beijing, and the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing.

His research interests include computer vision, pattern recognition, and remote sensing image processing, especially on object detection.



Kun Fu (Member, IEEE) received the B.Sc., M.Sc., and Ph.D. degrees from the National University of Defense Technology, Changsha, China, in 1995, 1999, and 2002, respectively.

He is currently a Professor with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China. His research interests include computer vision, remote sensing image understanding, and geospatial data mining and visualization.

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